LO6-Enhanced Entity-Relationship Modelling (Part 2)

Chapter 8 in Edition 6 and Chapter 4 in Edition 7

19 February 2026

Recap-Constraints on Specialisation/Generalisation Hierarchies

- **Multiple Specializations:**
  - A single superclass can have **multiple specializations**, each based on different distinguishing characteristics (refer to Figure 4.1).
- **Specialization with One Subclass:**
  - Some specializations may only have **one subclass**, like "MANAGER" in Figure 4.1.
  - In such cases, the circle notation is not used.

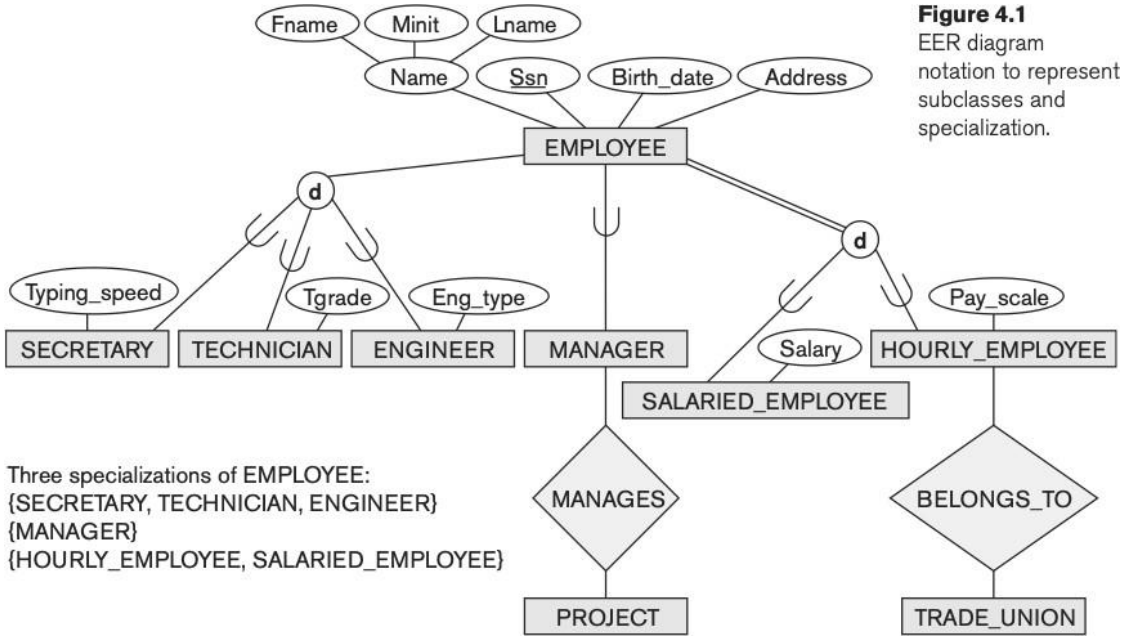


Figure 4.1
EER diagram notation to represent subclasses and specialization.

Constraints on Specialisation/Generalisation Hierarchies

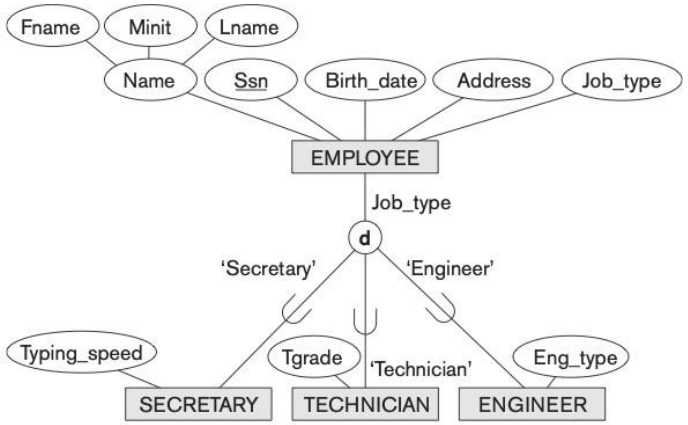


Figure 4.4
EER diagram notation for an attribute-defined specialization on Job_type.

Define **predicate-defined subclass**
Define an attribute-defined specialisation?

- Placing a constraint on the value of some member of the superclass will allow us to specify exactly which entities will become members of the subclass.
- These subclasses are referred to as predicate-defined (or condition-defined subclasses).
- Job_type = `Secretary', is called the defining predicate and Job_type in EMPLOYEE, the defining attribute.
- The specialisation is termed an attribute-defined specialisation.

Constraints on Specialisation/Generalisation Hierarchies

- Other constraints that may apply to a specialization
  - **Disjointness** (or disjointedness)
  - **Completeness** (or totalness)
- Disjointness and Completeness are independent and therefore the following combinations are possible:
  - Disjoint and Total
  - Disjoint and Partial
  - Overlapping and Total
  - Overlapping and Partial

Constraints on Specialisation/Generalisation Hierarchies

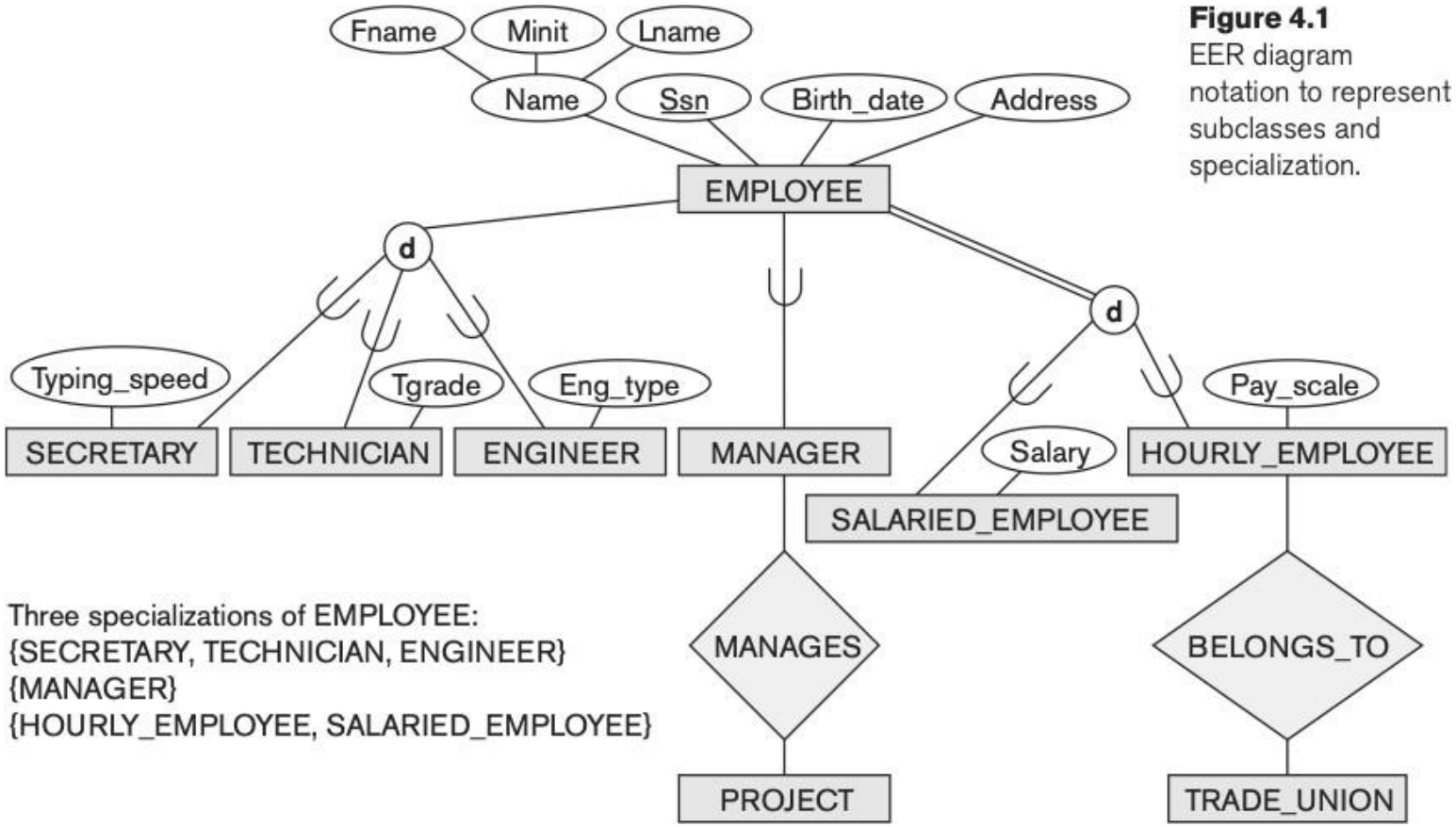


Figure 4.1
EER diagram notation to represent subclasses and specialization.

Three specializations of EMPLOYEE:
{SECRETARY, TECHNICIAN, ENGINEER}
{MANAGER}
{HOURLY_EMPLOYEE, SALARIED_EMPLOYEE}

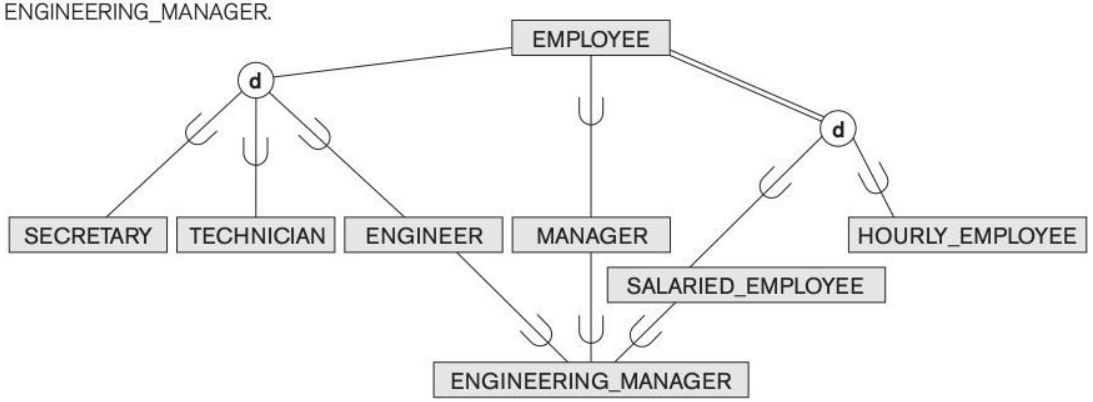
Formal Definitions for EER concepts

- A class is a set or collection of entities of a defined type.
- A subclass S is a class whose entities must always be a subset of the entities in another class, called the superclass C. S has an is-a relationship with C.
- A specialisation $Z = \{S_1, S_2, \ldots, S_n\}$ is a set of subclasses that have the same superclass G.
  - Z is said to be total if the superclass G is always equal to the union of all the subclasses. Otherwise it is partial.
  - Z is said to be disjoint if there is never an intersection between the subclasses. Otherwise it is overlapping.

Specialisation and Generalisation - Hierarchies and Lattices

- A subclass may have further subclasses. This nesting will either result in a tree-like (hierarchical) or graph-based (lattice) structure.
  - In a hierarchical structure, each subclass only has one parent.
  - In a lattice structure, each subclass can participate in more than one class/subclass relationship.

Figure 4.6
A specialization lattice with shared subclass ENGINEERING_MANAGER.



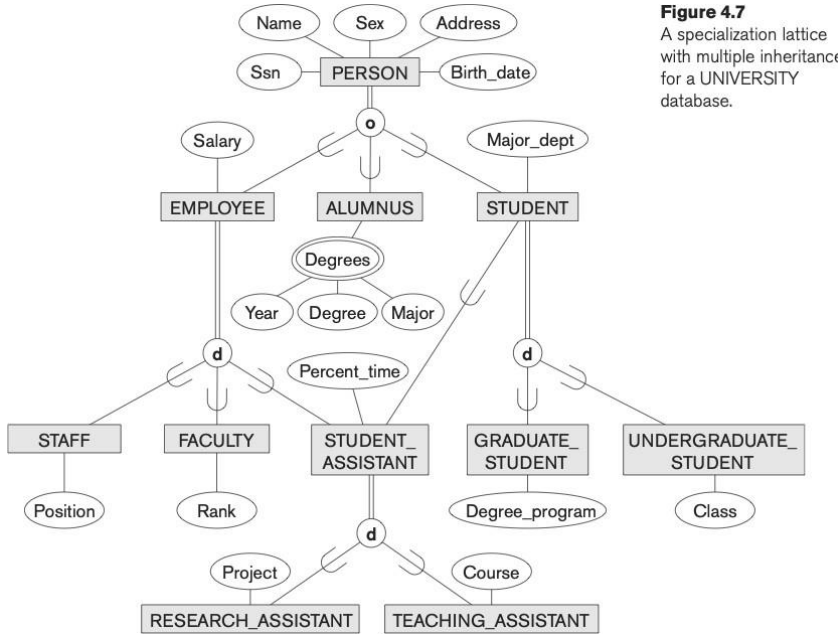


Figure 4.7
A specialization lattice with multiple inheritance for a UNIVERSITY database.

Modelling UNION Types

- Up till now we have considered entities with a single superclass. For example the EMPLOYEE entity type.
- Sometimes a superclass/subclass relationship requires more than one superclass.

Modelling UNION Types

- The subclass represents a collection of objects formed by the UNION of distinct entity types.
- The subclass is referred to as the **union type** (or category).
- A UNION is denoted in EER-diagrams with by placing a U in the circle.

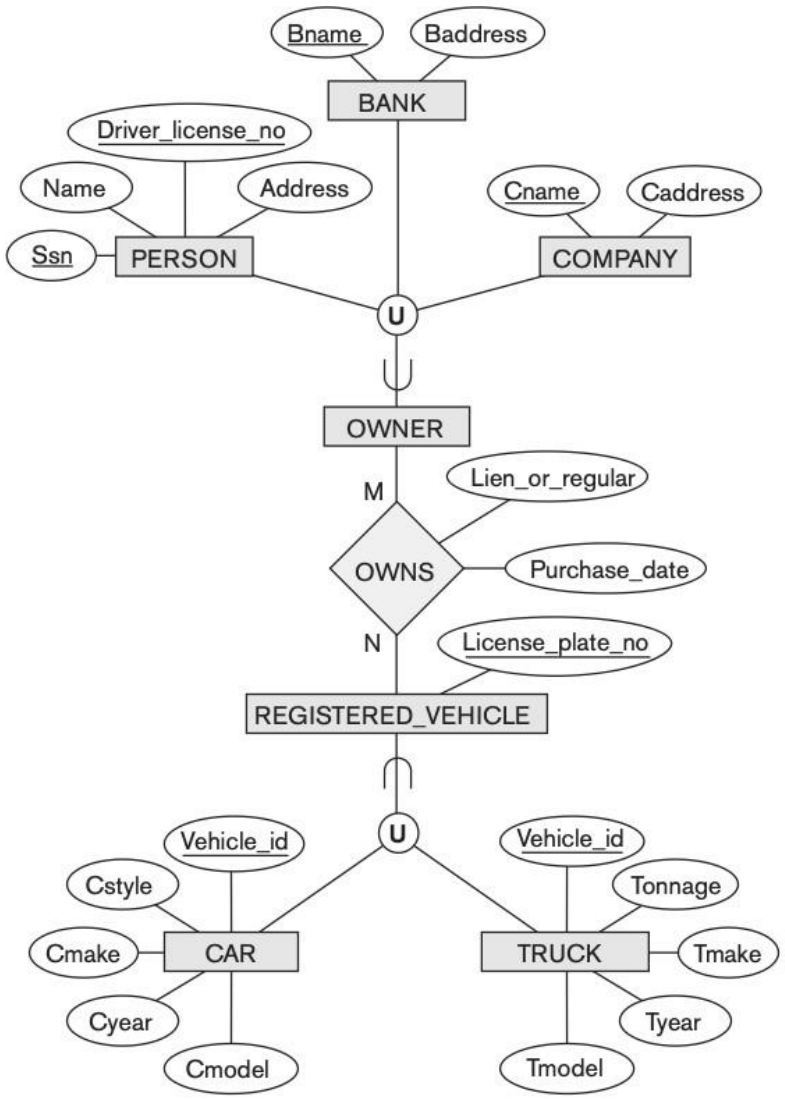


Figure 4.8
Two categories (union types): OWNER and REGISTERED_VEHICLE.

Modelling Union Types

- OWNER is a union type (or category) of the PERSON or BANK or COMPANY entity types. REGISTERED_VEHICLE is a subclass of CAR or TRUCK.
- An entity that is a member of OWNER, must exist in only one of its superclasses. That is, an OWNER may be a PERSON, BANK or COMPANY.
- Attribute inheritance works selectively for union types. The subclass inherits attributes from one of its superclasses. This means that a superclass of a category may hold different keys. For example, OWNER with PERSON, BANK or COMPANY holds different keys.

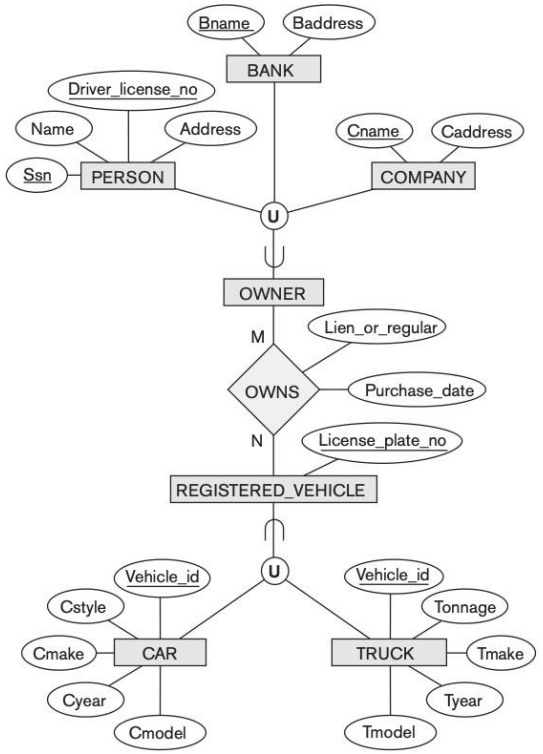


Figure 4.8
Two categories (union types): OWNER and REGISTERED_VEHICLE.

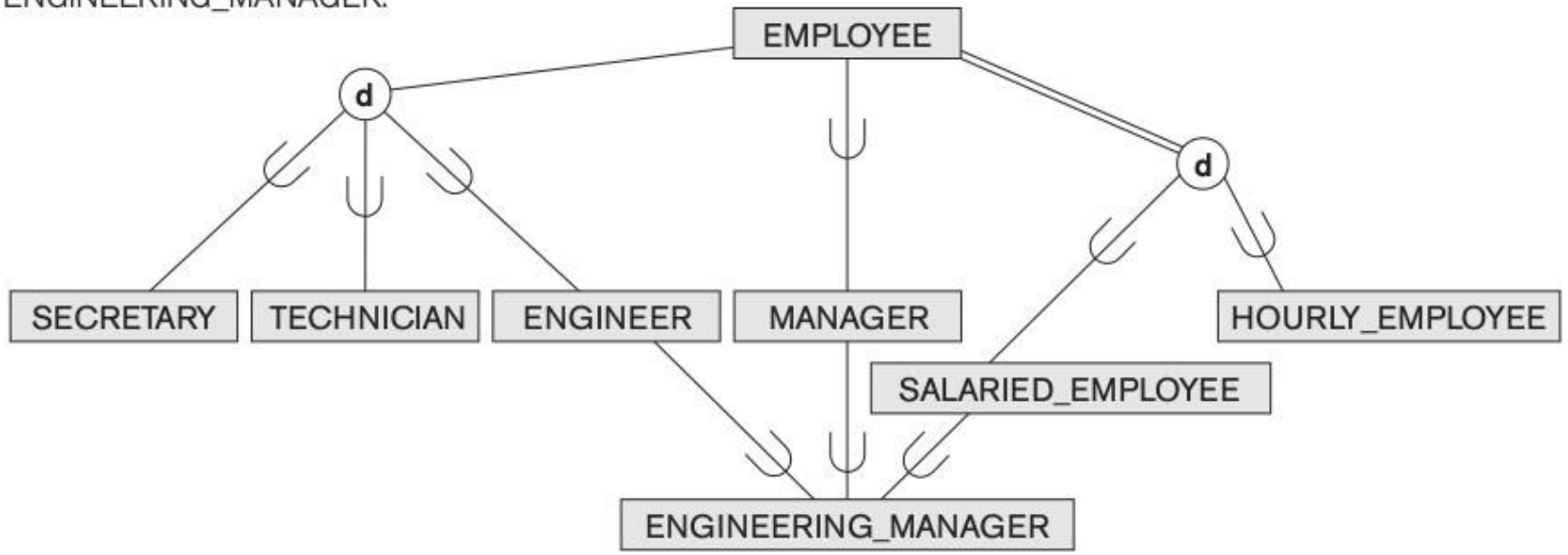
Modelling Union Types

- A union type may be partial or total.
  - A partial category holds a subset of the union.
  - A total category holds the UNION of all entities in its superclass. A double line connecting the category and the circle is used to differentiate the total category.

Modelling UNION Types

- Constrast a UNION with the engineering manager. An ENGINEERING_MANAGER must be an ENGINEER and a MANAGER and a SALARIED_EMPLOYEE.

Figure 4.6
A specialization lattice with shared subclass ENGINEERING_MANAGER.



EER Design Choices

- Specialisation and subclasses make the model more accurate but increase clutter.
- A subclass with few local attributes and no relationship types can be merged into superclass. Could then use `Type' attribute and `NULL' values for the non-members of the merged subclass.
- If all subclasses meet the requirements above, they can similarly be merged into their superclass.
- Avoid UNION types and categories as much as possible. Specialisation/generalisation usually sufficient.
- If the requirements do not indicate any particular constraints, the default would generally be overlapping and partial, since this does not specify any restrictions on subclass membership.

University Example - Requirements

- The database keeps track of three types of persons: employees, alumni, and students. A person can belong to one, two, or all three of these types. Each person has a name, SSN, sex, address, and birth date.
- Every employee has a salary, and there are three types of employees: faculty, staff, and student assistants. Each employee belongs to exactly one of these types. For each alumnus, a record of the degree or degrees that he or she earned at the university is kept, including the name of the degree, the year granted, and the major department. Each student has a major department.
- Each faculty has a rank, whereas each staff member has a staff position. Student assistants are classified further as either research assistants or teaching assistants, and the percent of time that they work is recorded in the database. Research assistants have their research project stored, whereas teaching assistants have the current course they work on.
- Students are further classified as either graduate or undergraduate, with the specific attributes degree program (M.S., Ph.D., M.B.A., and so on) for graduate students and class (freshman, sophomore, and so on) for undergraduates.

University Example - Notations

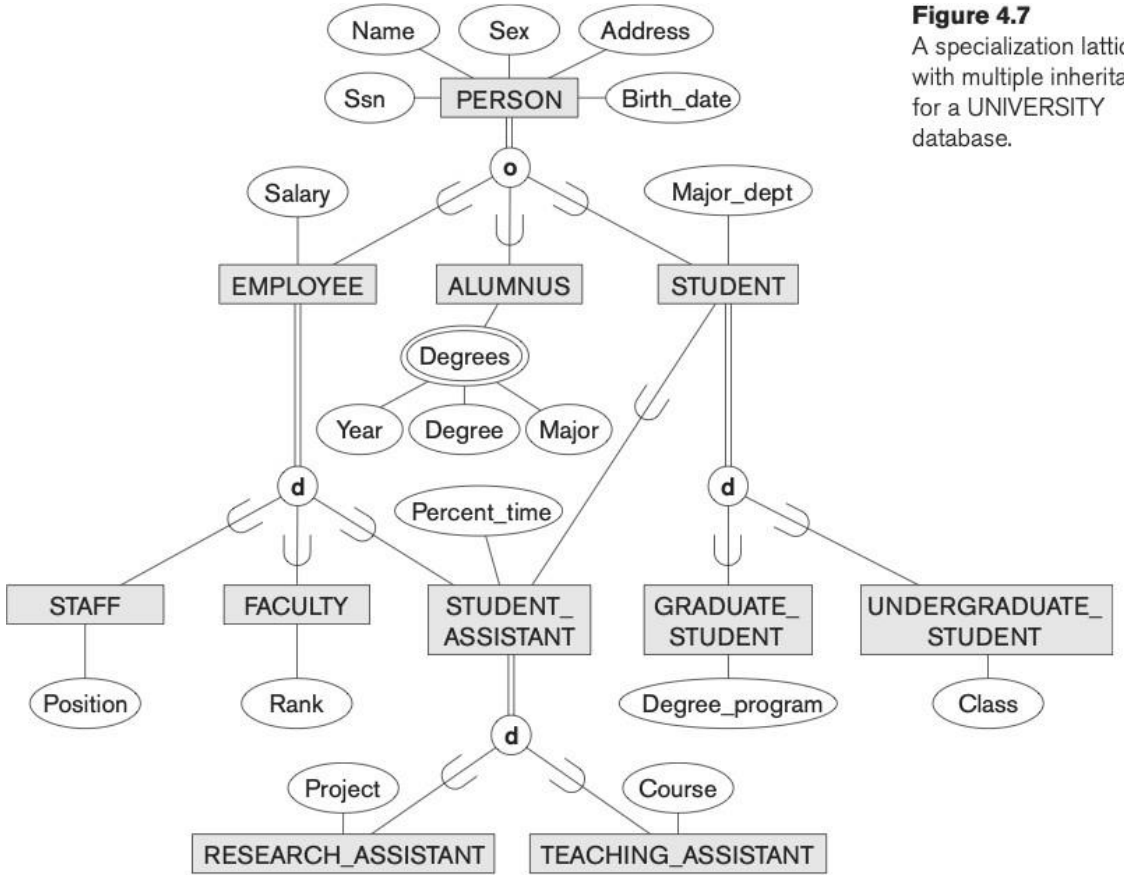


Figure 4.7
A specialization lattice with multiple inheritance for a UNIVERSITY database.

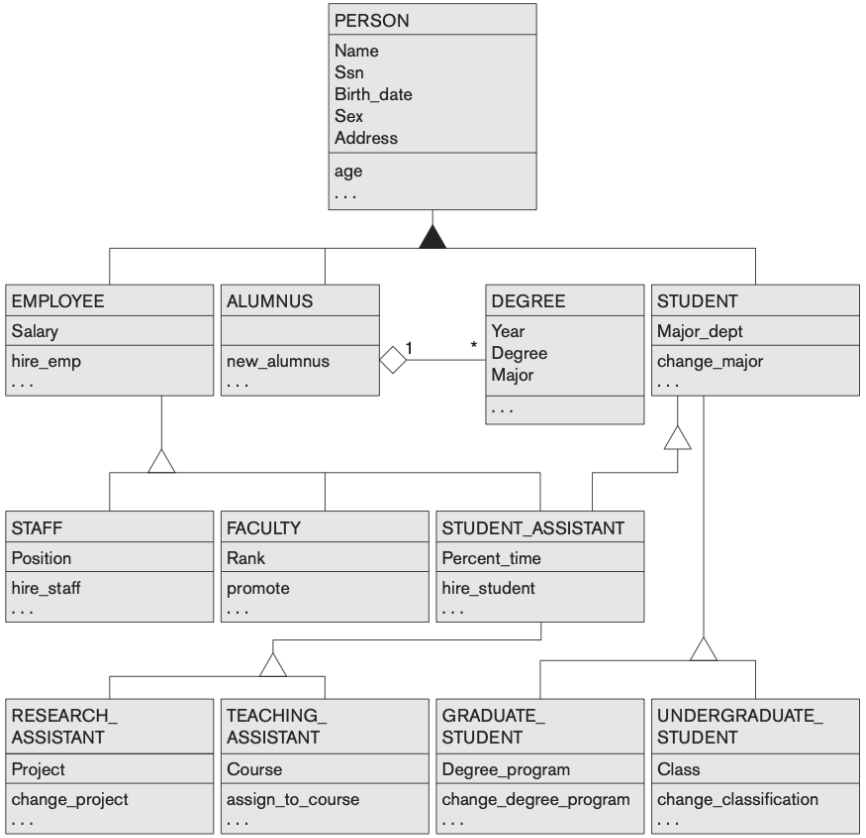


Figure 4.10
A UML class diagram corresponding to the EER diagram in Figure 4.7, illustrating UML notation for specialization/generalization.

Exercise:

- Consider a GRADE_BOOK database in which instructors within an academic department record points earned by individual students in their classes. The data requirements are summarized as follows:
- Each student is identified by a unique identifier, first and last name, and an e-mail address.
- Each instructor teaches certain courses each term. Each course is identified by a course number, a section number, and the term in which it is taught. For each course he or she teaches, the instructor specifies the minimum number of points required in order to earn letter grades A, B, C, D, and F. For example,90 points for an A, 80 points for a B, 70 points for a C, and so forth.
- Students are enrolled in each course taught by the instructor.
- Each course has a number of grading components (such as midterm exam, final exam, project, and so forth). Each grading component has a maximum number of points (such as 100 or 50) and a weight (such as20% or 10%). The weights of all the grading components of a course usually total 100.
- Finally, the instructor records the points earned by each student in each of the grading components in each of the courses. For example, student *1234* earns 84 points for the midterm exam grading component of the section 2 course CSc2310 in the fall term of 2009. The midterm exam grading component may have been defined to have a maximum of 100 points and a weight of 20% of the course grade.
- Design an enhanced entity–relationship diagram for the grade book database..